The Alternative & Augmentative Communication / Cortical Visual Impairment Matrix

Alternative & Augmentative Communication (AAC)/
Cortical Visual Impairment (CVI) Matrix:
Student-Centered Guidelines for AAC & Expressive Communication Development

An instrument for professionals and families to balance the results of Communication Matrix assessment (Rowland, 1996; Rev. 2004) with CVI Range assessment (Roman, 2007; Rev. 2018)

Developed by Christopher Russell & Jennifer Willis

NEW YORK DEAF-BLIND COLLABORATIVE

Christopher Russell, MS. Ed., TVI,
CVI Range Endorsed
Project Coordinator
New York Deaf-Blind Collaborative
christopher.russell@qc.cuny.edu

Jennifer Willis, M.S., ToD/HH,
CVI Range Endorsed
Project Coordinator
Connections Beyond Sight and Sound
jcwillis@umd.edu

This instrument was developed using content from:

Roman, C. (2018) Cortical visual impairment: An approach to assessment and intervention (2nd Ed.). New York, NY: AFB Press.

Rowland, C. (1996; Rev. 2004). Communication matrix. Portland, OR: Design to Learn. http://www.communicationmatrix.org

Chris Russell, New York Deaf-Blind Collaborative: christopher.russell@qc.cuny.edu
Jennifer Willis, Connections Beyond Sight and Sound: jcwillis@umd.edu

AAC: Context and Challenges

- Significant mismatches between communication systems and visual access:
  - Communication that is visually inaccessible
  - Communication that is visually accessible, but inappropriate in terms of communication development (current expressive levels)

  - Phase I example: eye gaze system
  - Phase III example: complex 2D iPad system, child is currently a pre-symbolic communicator

A Balanced Communication Plan

Sensory Access

- CVI Phase and Characteristics
- Preferred learning channels

Communication Level Access

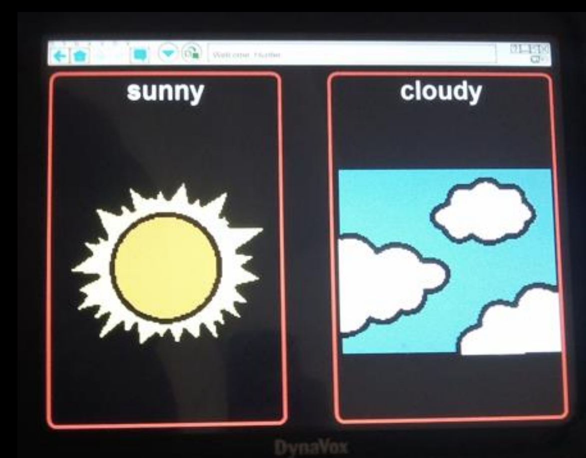
- Presymbolic/Symbolic
- Prelinguistic/Linguistic

- Multi-sensory access
- Universal Design

- Multimodal communication
- Provide a robust AAC program receptively

Multiple Modes / Multiple Systems

- Aided
- Unaided
- No-tech
- Light-tech
- High-tech



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NYDBC
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CONNECTIONS BEYOND SIGHT AND SOUND

Christopher Russell, MS. Ed., TVI,
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Project Coordinator
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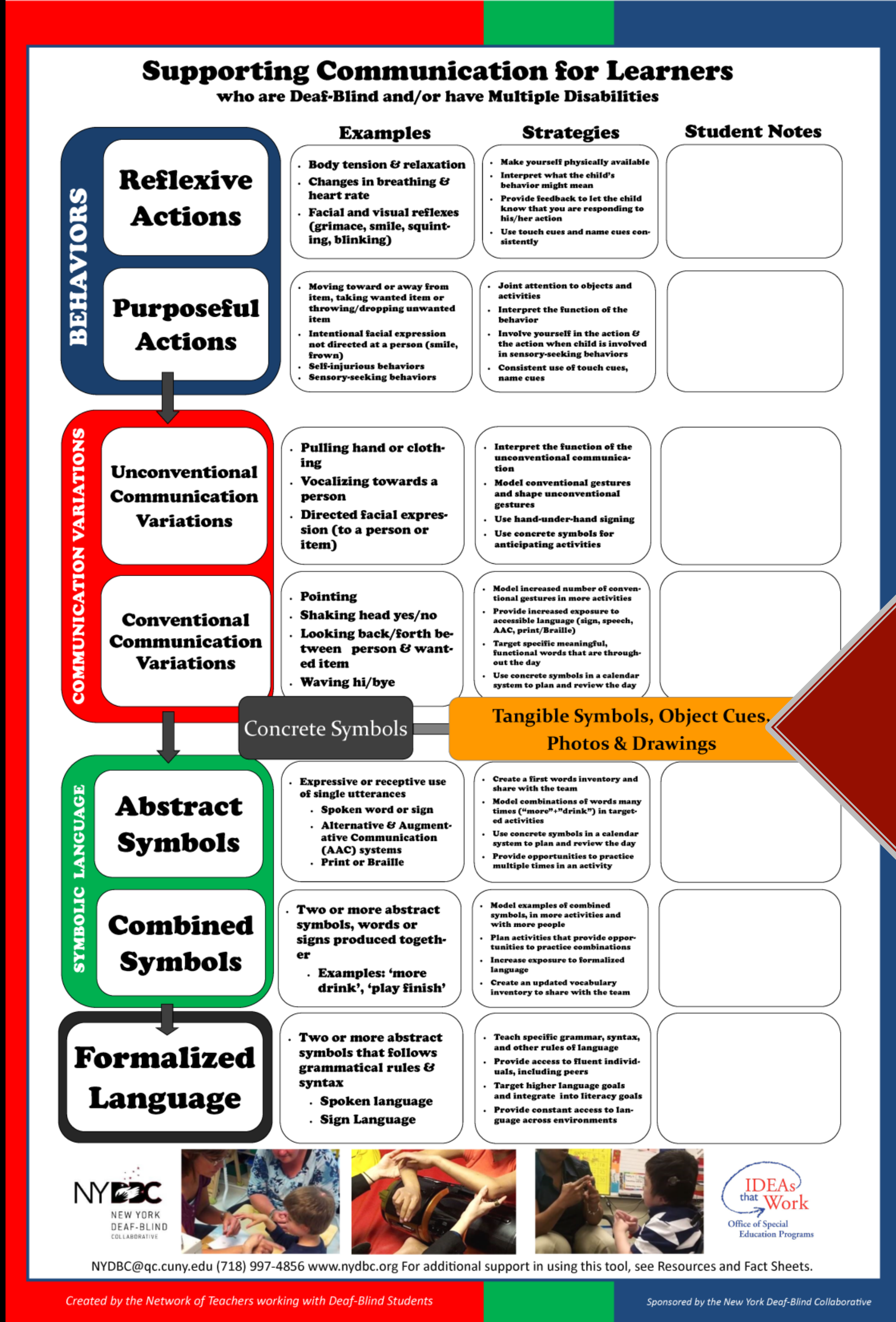


Supporting Communication for Learners

who are Deaf-Blind and/or have Multiple Disabilities

		Examples	Strategies	Student Notes
BEHAVIORS	Reflexive Actions	• Body tension & relaxation • Changes in breathing & heart rate • Facial and visual reflexes (grimace, smile, squint-ing, blinking)	• Make yourself physically available • Interpret what the child's behavior might mean • Provide feedback to let the child know that you are responding to his/her action • Use touch cues and name cues con-sistently	
BEHAVIORS	Purposeful Actions	• Moving toward or away from item, taking wanted item or throwing/dropping unwanted item • Intentional facial expression not directed at a person (smile, frown) • Self-injurious behaviors • Sensory-seeking behaviors	• Joint attention to objects and activities • Interpret the function of the behavior • Involve yourself in the action & the action when child is involved in sensory-seeking behaviors • Consistent use of touch cues, name cues	
COMMUNICATION VARIATIONS	Unconventional Communication Variations	• Pulling hand or cloth-ing • Vocalizing towards a person • Directed facial expres-sion (to a person or item)	• Interpret the function of the unconventional communica-tion • Model conventional gestures and shape unconventional gestures • Use hand-under-hand signing • Use concrete symbols for anticipating activities	
COMMUNICATION VARIATIONS	Conventional Communication Variations	• Pointing • Shaking head yes/no • Looking back/forth be-tween person & want-ed item • Waving hi/bye	• Model increased number of conven-tional gestures in more activities • Provide increased exposure to accessible language (sign, speech, AAC, print/Braille) • Target specific meaningful, functional words that are through-out the day • Use concrete symbols in a calendar system to plan and review the day	
	Concrete Symbols	Tangible Symbols, Object Cues, Photos & Drawings		

SYMBOLIC LANGUAGE	Abstract Symbols	• Expressive or receptive use of single utterances
		  • Spoken word or sign
		  • Alternative & Augment-ative Communication (AAC) systems
		  • Print or Braille
			• Create a first words inventory and share with the team
			• Model combinations of words many times ("more"+"drink") in target-ed activities
			• Use concrete symbols in a calendar system to plan and review the day
			• Provide opportunities to practice multiple times in an activity	

SYMBOLIC LANGUAGE	Combined Symbols	• Two or more abstract symbols, words or signs produced togeth-er • Examples: 'more drink', 'play finish'	• Model examples of combined symbols, in more activities and with more people • Plan activities that provide oppor-tunities to practice combinations • Increase exposure to formalized language • Create an updated vocabulary inventory to share with the team	

	Formalized Language	• Two or more abstract symbols that follows grammatical rules & syntax
		  • Spoken language
		  • Sign Language
			• Teach specific grammar, syntax, and other rules of language
			• Provide access to fluent individ-uals, including peers
			• Target higher language goals and integrate into literacy goals
			• Provide constant access to lan-guage across environments	

NYDBC NEW YORK DEAF-BLIND COLLABORATIVE

IDEAs that Work Office of Special Education Programs

NYDBC@qc.cuny.edu (718) 997-4856 www.nydbc.org For additional support in using this tool, see Resources and Fact Sheets.

Created by the Network of Teachers working with Deaf-Blind Students

Sponsored by the New York Deaf-Blind Collaborative

CONCRETE MODES OF COMMUNICATION:

The bridge from early communication (behaviors) to first words and language

Tangible Symbols (Transitional Concrete)

Can be:

- whole objects
- parts of objects
- associated objects
- textures or shapes
- line drawings
- Photographs

Consider:

level of iconicity / abstraction

Does the student have *symbolism*?

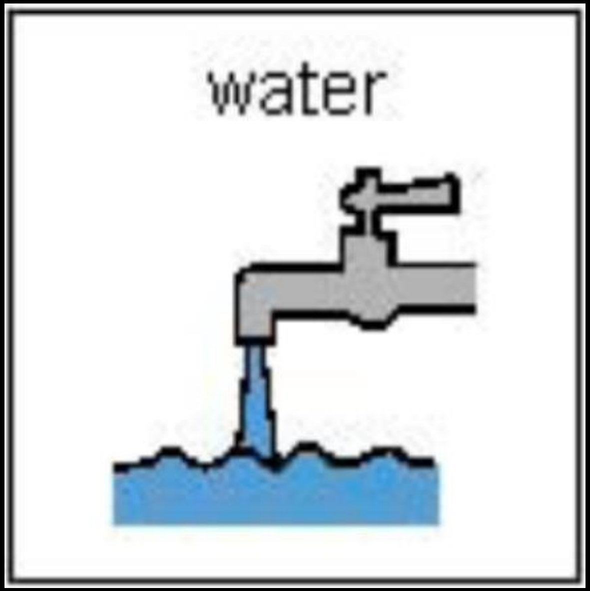
- The understanding that a symbol refers to an event/activity, object, person

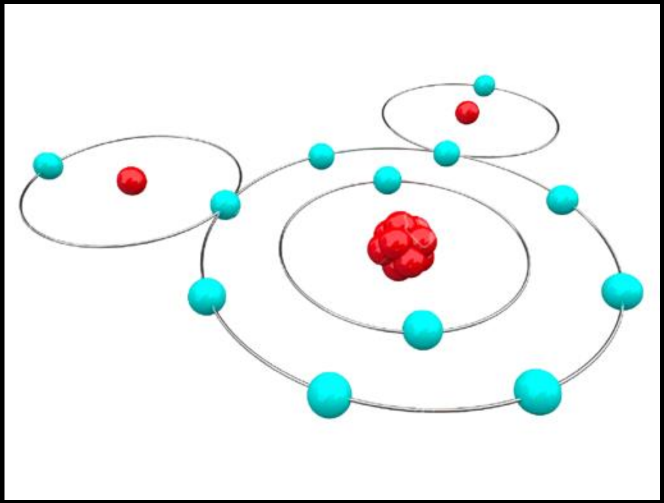
SYMBOL → REFERENT

ICONICITY – Conceptual & Visual Complexity

(CONCRETE → ABSTRACT)

ICON → INDEX → SYMBOL





Is this your experience with water?

Iconicity (like novelty) is relative to individual experience.

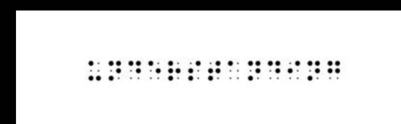
Symbol Hierarchy

Whole/identical
object

Partial/Asscoiated
object

Photo/ Picture
Symbol
Tangible Symbol

shoe



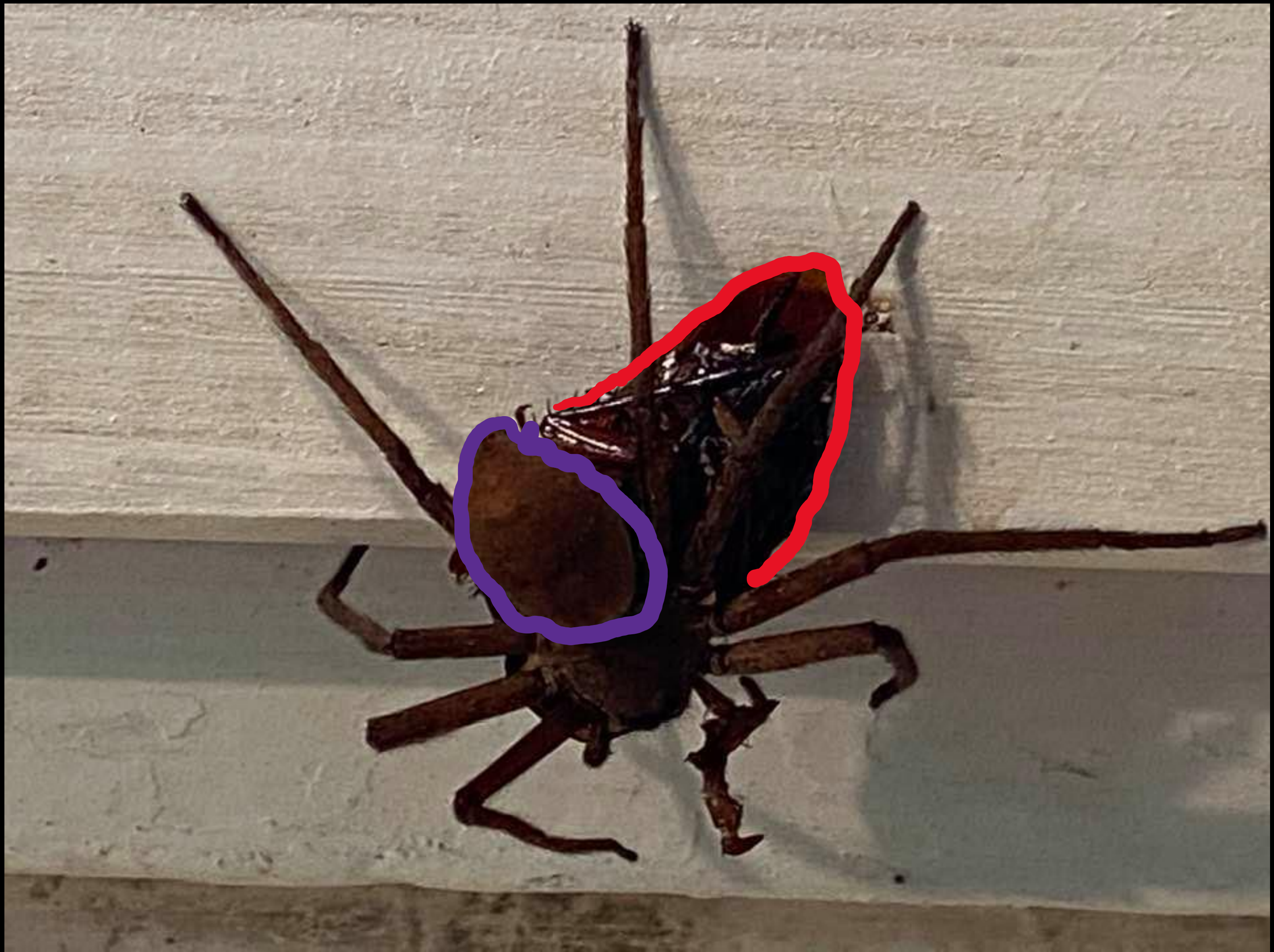
Print/ Braille

Spoken words

Signs


Core Words
like
break
All Word Lists
home page
help
STOP
stop
bathroom
Topics
go
more
all done
ABC
Keyboard

Even photos can be visually/conceptually complex…



Uses of Tangible Cues/Symbols - Levels

1) Anticipation of an activity

- To learn symbolism – connection between a symbol and referent
- Reducing stress in transitions

2) Transitional communication systems (more complex uses for children/youth who have symbolism

Tangible Symbols:
Clarifying Tactile Traditions

Traditionally used for students with ocular visual impairments, based on tactile properties, NOT visual

- Communication purpose is to bridge presymbolic and symbolic modes, develop symbolism
  - Provide a CONCRETE referent

- NEVER intended to be an end, rather a means to an end (symbolism)

Concrete Symbols: Considerations for CVI

Select based on visual properties, NOT tactile

- This is the opposite for ocular VI/tactile learners

Complexity:

symbolic and visual

- Adaptations – based on Range
  - 3D/2D, color, complexity, etc

- What are the CVI adaptations?
- What are the communication modifications?

CVI:

- Partial objects (not yet 2-D)
- Black backdrop (visual clutter)
- Red rectangle highlight
- Red tape on rim of finished box

Communication – Levels and Modes:

- Concrete (tangible) symbols
- Visual sign with tactile modifications
- Constant contact/tactile support
- Speech / audition
- Prompts / wait time

Systems that grow with the child

- From 3D to 2D
- From Pre-symbolic to Symbolic
- Developing complex language

Photos and Line Drawings

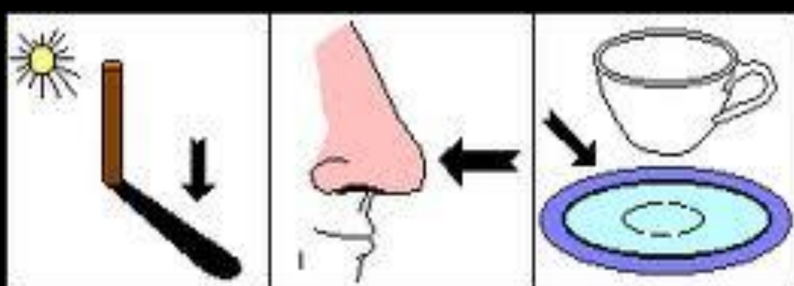
- Which do you think would be more abstract?

Who should use photos / line drawings?

- Enough vision?
- For CVI – Phase 2 or 3
- Understanding of what the photo / drawing represents

Visual components must also be **conceptually** appropriate

- Based on communication assessment
- Collaboration with team/SLP



?

Line drawings are very abstract!

http://www.pathstoliteracy.org/strategies/sequence-board-child-cvi

Sequence Board for my son with CVI

Case Study: Henry (mid-Phase II on CVI Range)
Shared courtesy of Rachel Bennett

- Henry’s able to attend to very familiar 2D pictures on the iPad, but it is a challenging visual processing task for him so he easily gets visually fatigued.

- His tactile calendar system allows him access and to stay anchored throughout the day no matter his visual fatigue or how he’s feeling.

- Recently, he’s been able to recognize a 2D image of familiar icons on his iPad presented in an array of 1-2.

Sample from morning routine (from left to right):

Meal time (breakfast)

Change clothes

Brush teeth

From left to right:

Snack time (wrapper from one of his favorite snacks)

Reading time

Play time

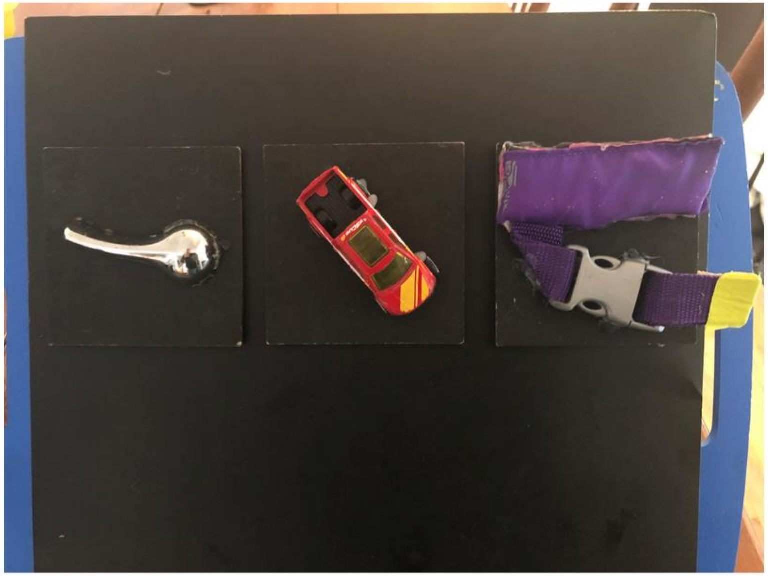
From left to right:

Henry

Walks

Bakery

**His schedule icons can also be used to form a sentence about an upcoming experience. Henry walks to one of the three local bakeries almost everyday.



From left to right:

Bathroom time

Car ride

Pool (Piece of his old life jacket)

Phase I Intervention: Most Characteristics

Goal: Building visual behavior

Looking is a goal in itself

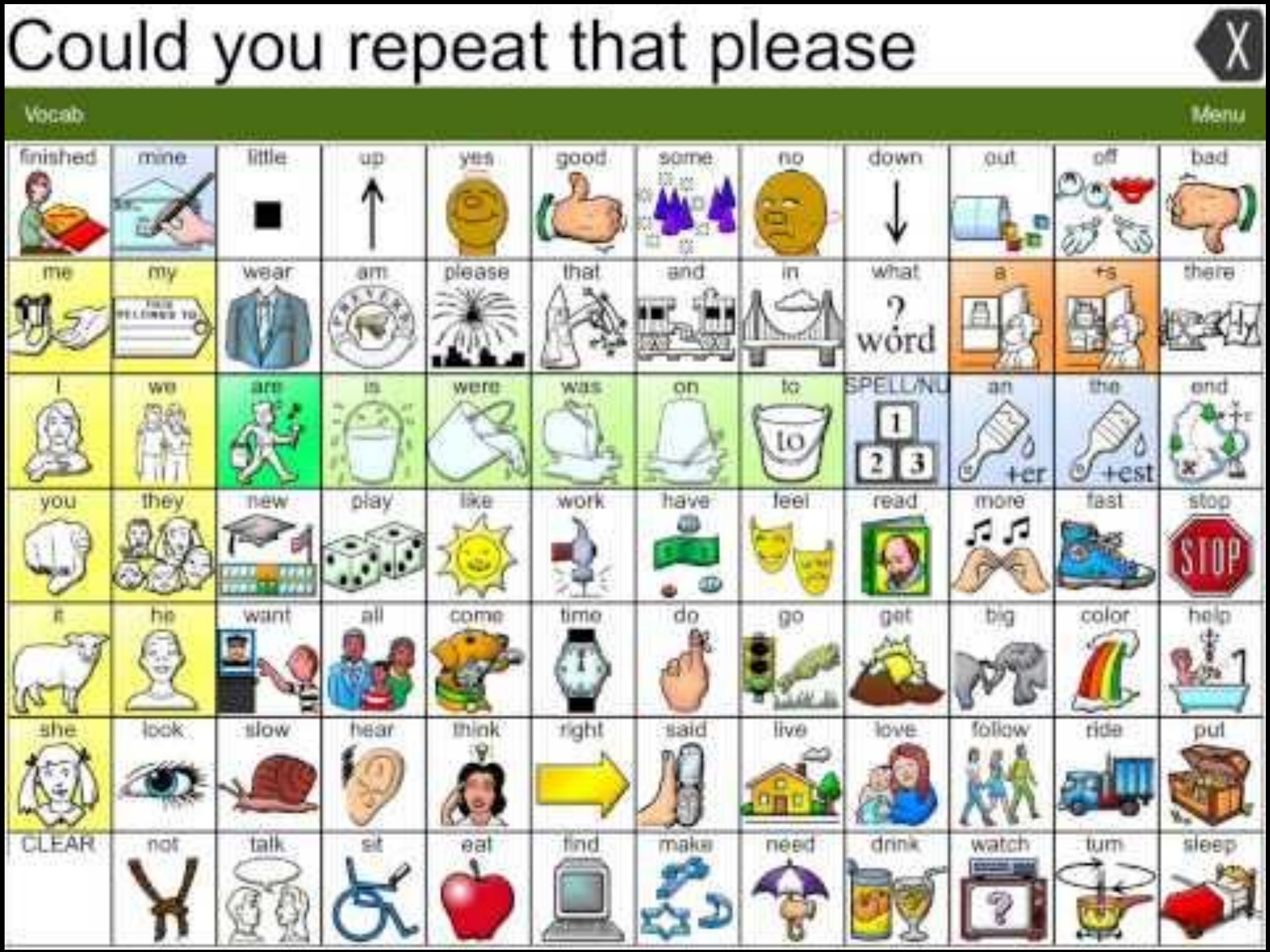
High level of environmental control

- NO visual processing of 2-D
- Maximize visual access to modalities but don't expect visual fixation

- Auditory scanning?
  - What if DB?
- Tactile components?
- MULTIMODAL

Tactile Adaptations

What about auditory scanning or motor memory access for Phase I?



Integrate accessible switches into self-directed visual experiences based on CVI Range during down-time

Multisensory overstimulation is observable… if you're paying attention.

Phase II Intervention

Goal: Integrate Vision and Function

- Able to use vision in activities, with adaptations and opportunities

- Early → Late Phase II
  - Level of visual adaptation needed

  - 2D emerging

What needs to be adapted visually in order to elicit and sustain visual attention at targeted points in a routine?

CVI Schedule & Planning Tools

Phase II & III and Language/Symbolism

You don’t need symbolism and language to reach Phase late II & Phase III (ventral stream processing)

but…

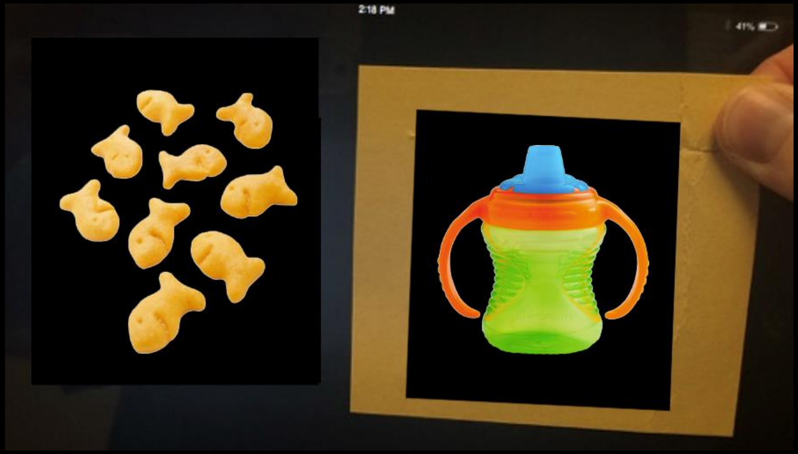
How do you identify, recognize and discriminate 2D symbols if you do not yet have symbolism & language?

Partner-Assisted Scanning

Pointing, showing, speaking messages a student will select

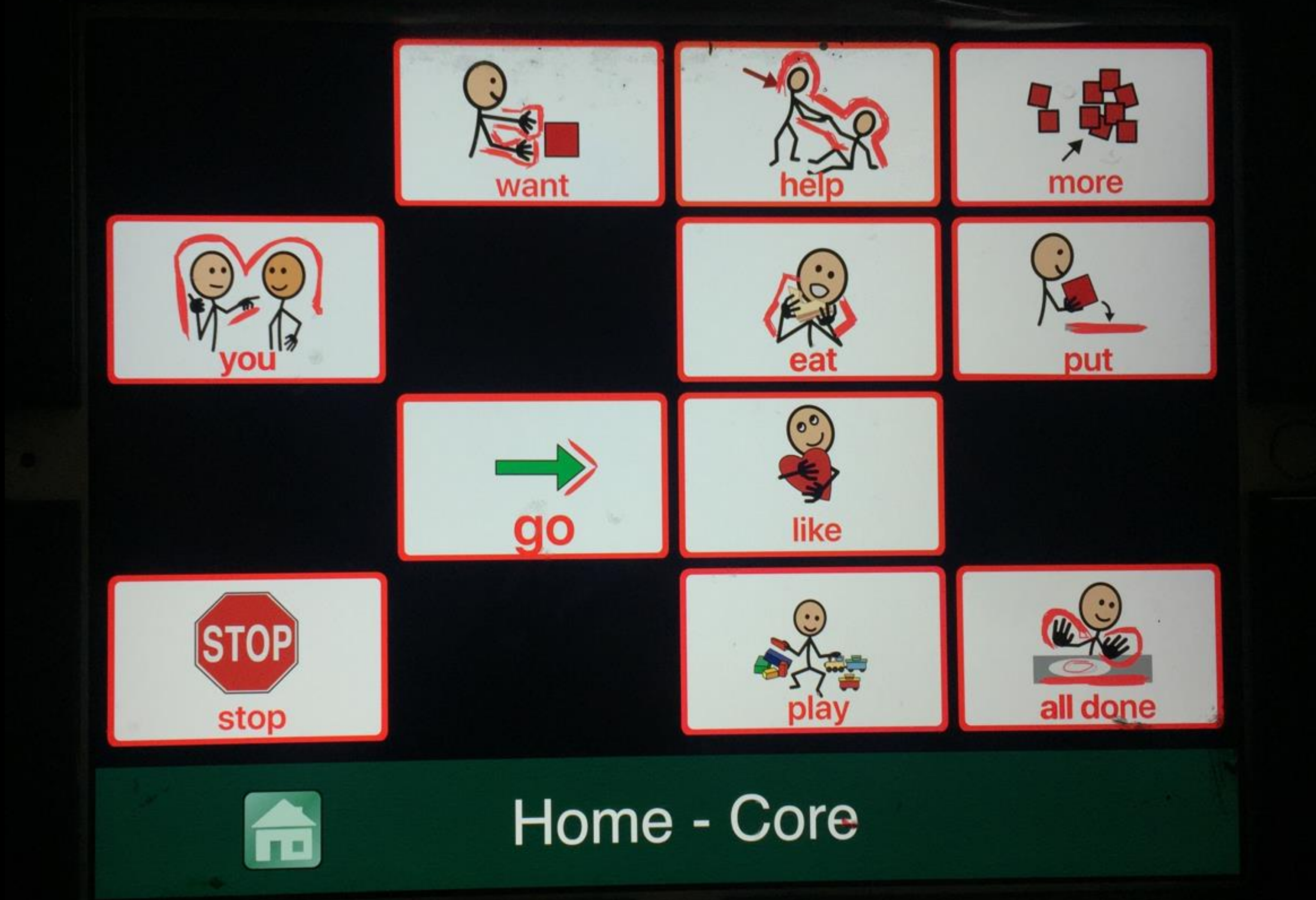
- Facilitate use of current receptive vocabulary
- Teach new symbols/words
- Develop visual skills
- Pair with other modes
- Expand: levels of selection
- Can be used across multiple forms

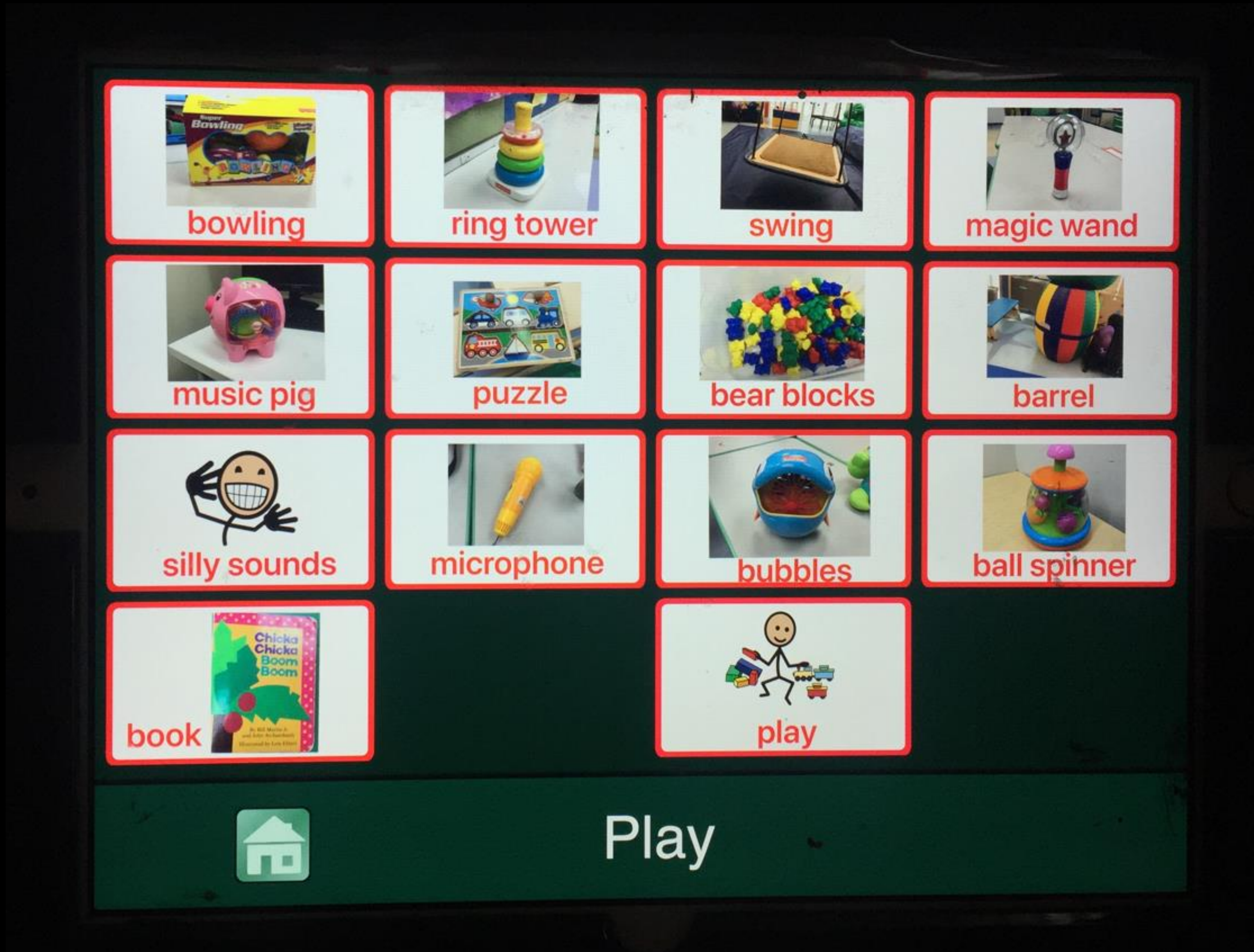
(Burkhart & Porter, 2012; Hanser, 2007)



Partner assisted scanning on an iPad with a yellow cardboard square cutout/guide

Reliant upon responsive communication partner


want
help
more
you
eat
put
go
like
stop
STOP
play
all done
Home - Core


bowling
ring tower
swing
magic wand
music pig
puzzle
bear blocks
barrel
silly sounds
microphone
bubbles
ball spinner
Chicka Chicka Boom Boom
book
play
Play

Don’t forget about visual fields!


let's eat
I want to drink
bathroom
RESTROOM

let's eat

Eye Gaze Systems: When are they appropriate?

- Eye to object contact
  (4-5 at earliest on Range)
- Visual localization skills
- Consider fatigue

- 2-D?
  - Depends on the use: communication system or visual exploration/games?

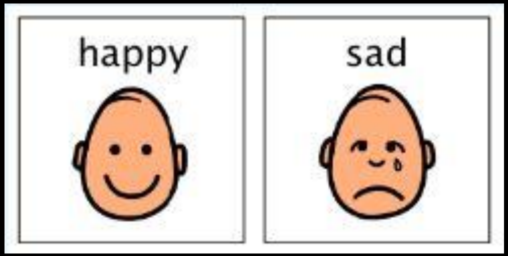
E Tran Board

Phase III Intervention

Goal: Refinement of the CVI Characteristics

- Demonstrate visual curiosity
- Can process 2-D
- Need adaptations to support learning and visual vocabulary

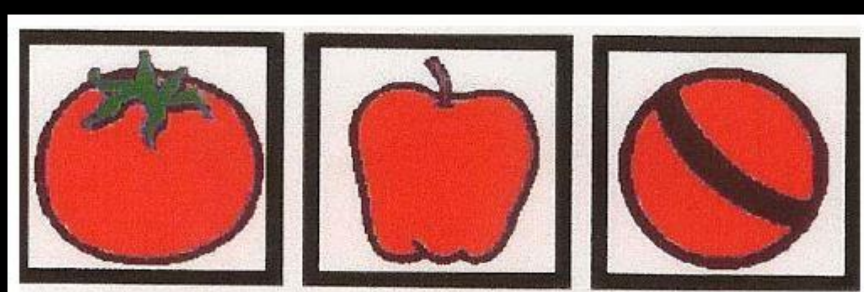
Salient feature - what specific part of the symbol gives it unique meaning?

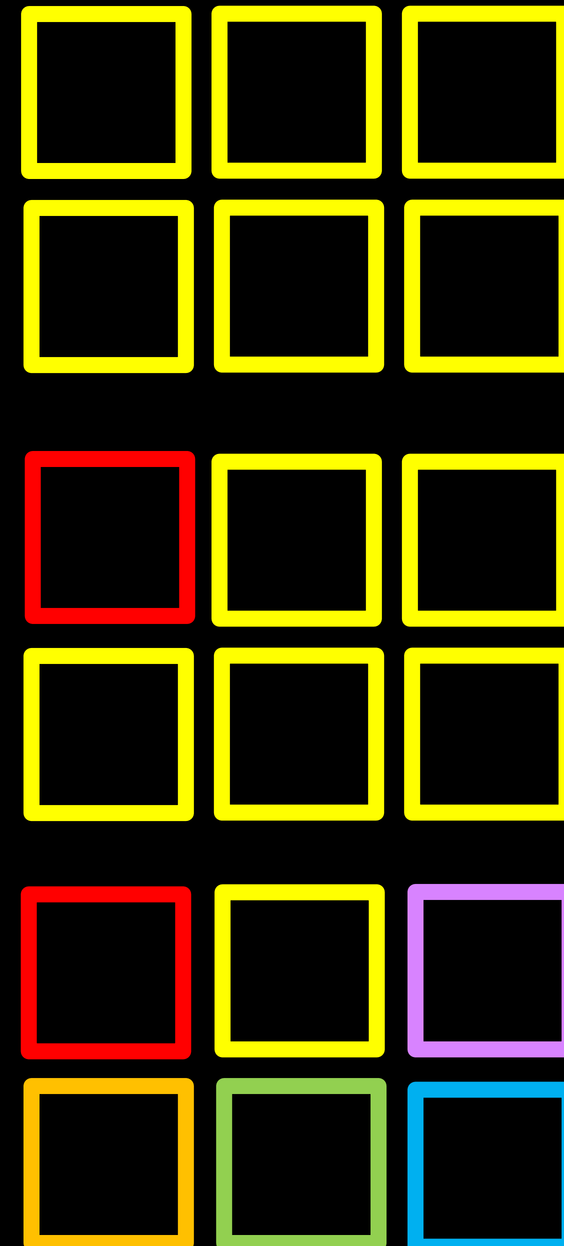


"Critical component" (Bent & Buckley, 2013)

Color Considerations

- Contrast / bright colors
- AVOID Pastels: difficult for CVI, optic nerve atrophy, optic nerve hypoplasia
- Vary the position of colors, use colors to stand out

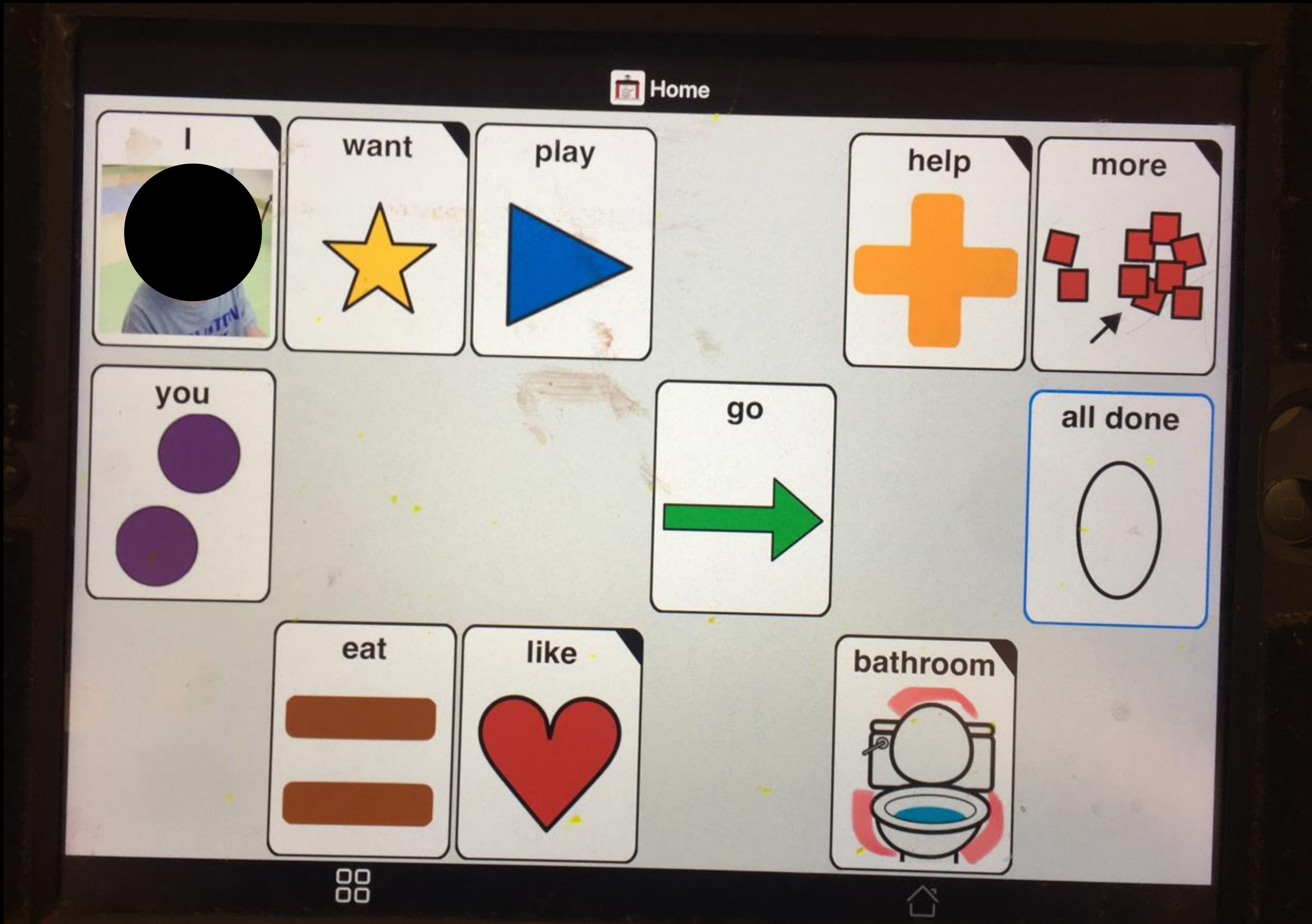


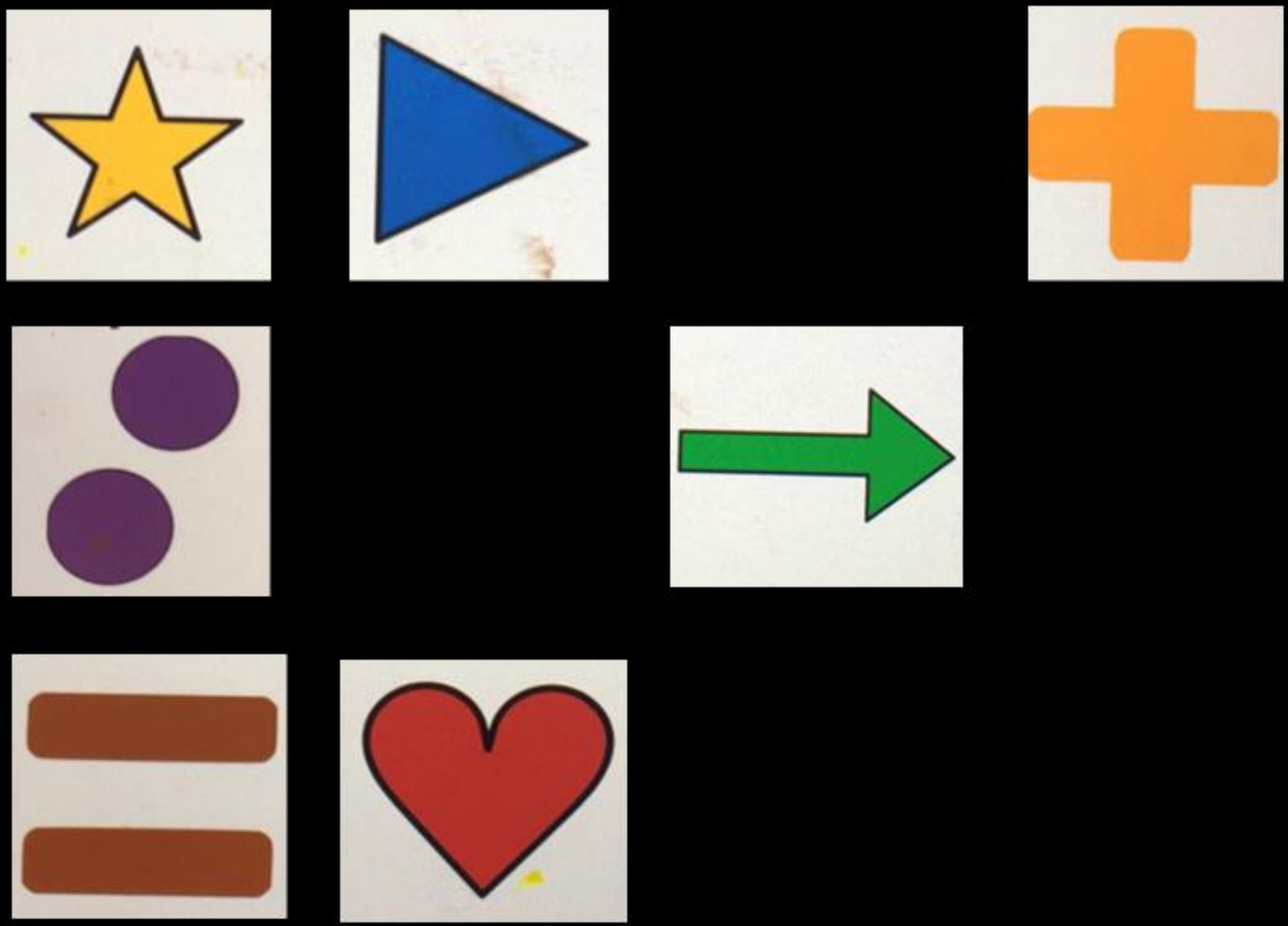


(Kreuzer & King, 2004; Kreuzer, 2007)

More Considerations for Symbols

- As conceptually concrete as the student needs
- Complexity of symbol & array
  - *What's the Complexity Framework* (Tietjen)
- Visually DISTINCT when placed alongside other symbols
  - Consider the student's discrimination, identification & recognition skills


Home
I
want
play
help
more
you
go
all done
eat
like
bathroom



want play help

you go

eat like

want play help

you go

like

eat

snack

applesauce

olives

cookies

Core Words

Go

Kitchen

Living room

Car

Outside

Inside

Emma's Talker

Chat

greetings

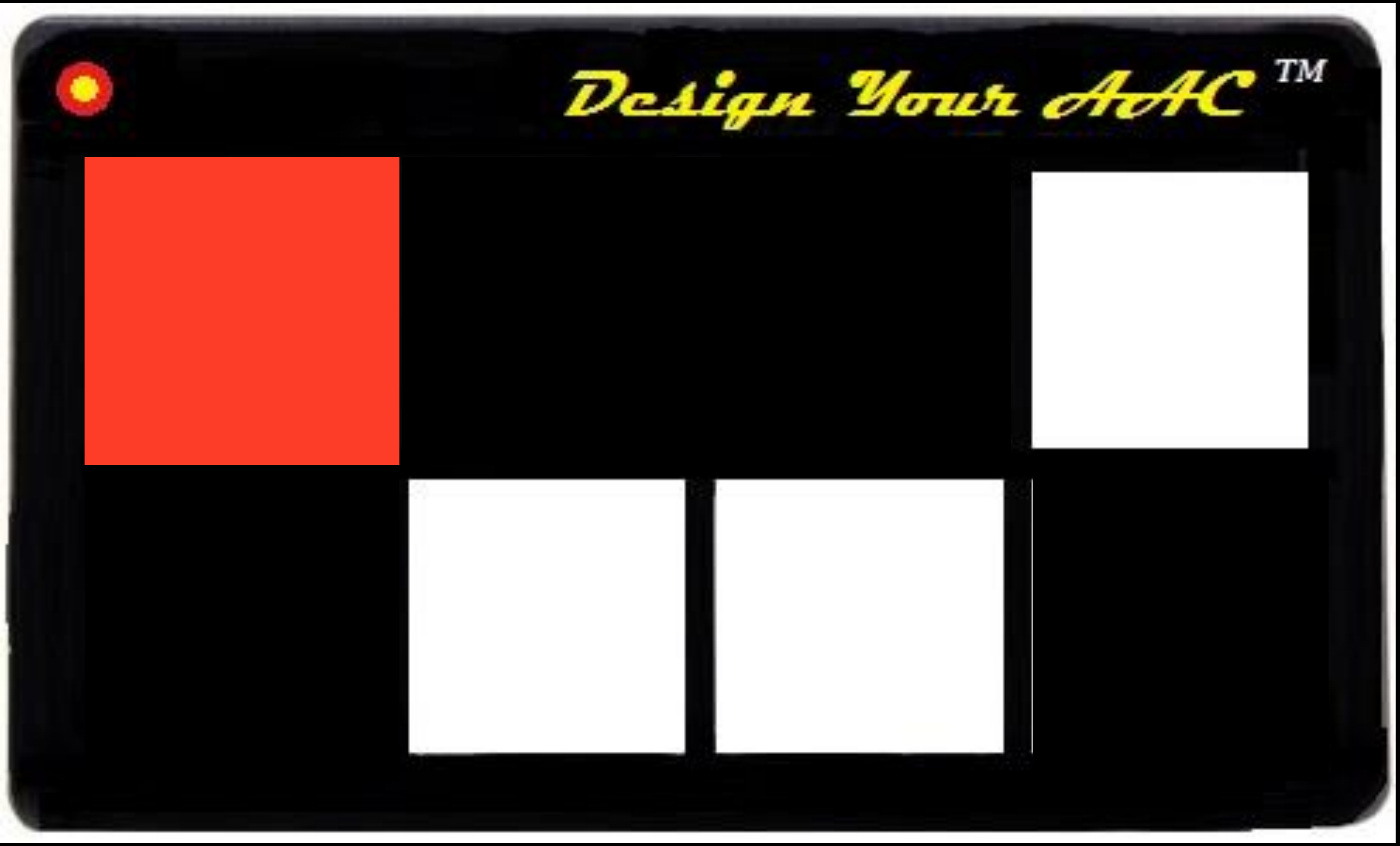
feeling

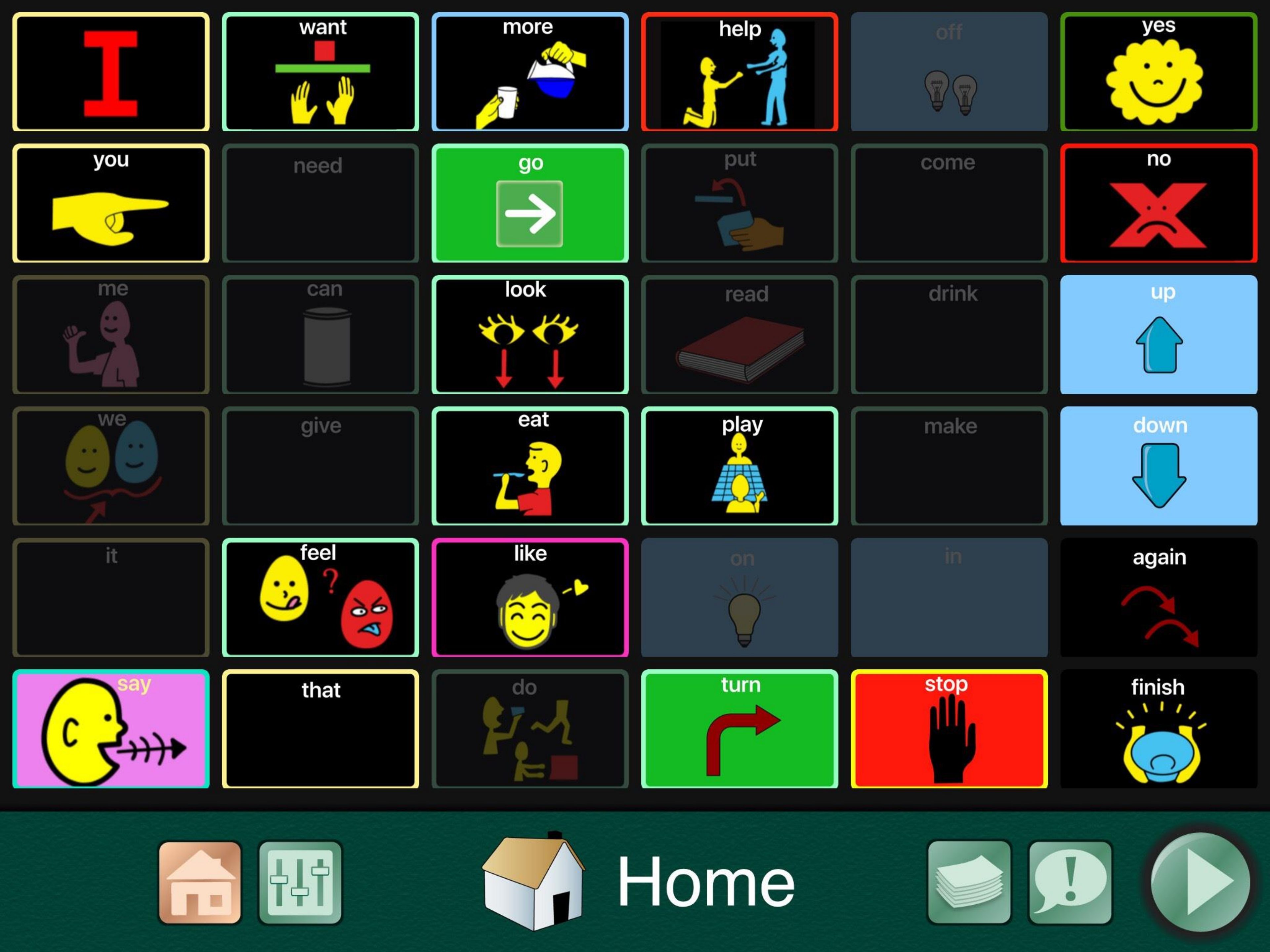
expressions

comments

questions

Don't limit the number of cells on a device;
Limit the number of cells used in the array





I
want
more
help
off
yes
you
need
go
put
come
no
me
can
look
read
drink
up
we
give
eat
play
make
down
it
feel
like
on
in
again
say
that
do
turn
stop
finish
Home

High tech or low tech?

“**Having an AAC device doesn’t make you a communicator any more than having a piano makes you a musician**.” – PraacticalAAC.org

Does the student know the symbols already?

Or is he/she learning them while using the AAC display?

Modeling Communication

Accessible modeling

- visual access
- processing time

Honor the child's existing mode(s) of communication

More on modeling AAC

Social modeling – will peers be communicating too?

- Possibility of a level on the display for when the student is fatigued or most impacted by multisensory complexity *(credit to Lynn Elko)*

Make a low-tech backup!

Salient Feature Instruction of Symbols

- Can this be done in the context of communication routines with the symbols?
- Does it need to be taught "on the side"?
  - With printed versions of symbols, on a separate display?
  - Do these salient features then need to be incorporated onto the symbols on the AAC device?

Parent Example: "Explain Everything" app

Specific High-Tech AAC Programs

Proloquo2Go
Snap + Core
GoTalk Now
TouchChat
Speak For Yourself
LAMP
Others…

The program is not the most important thing… but some programs are less adaptable than others.

Consider potential for:

- Modifications to layout and array
- Ability to mask cells
- Modifications to individual symbols
- Ability to use photos
- Output display (auditory, visual text)

Consider adaptations to…

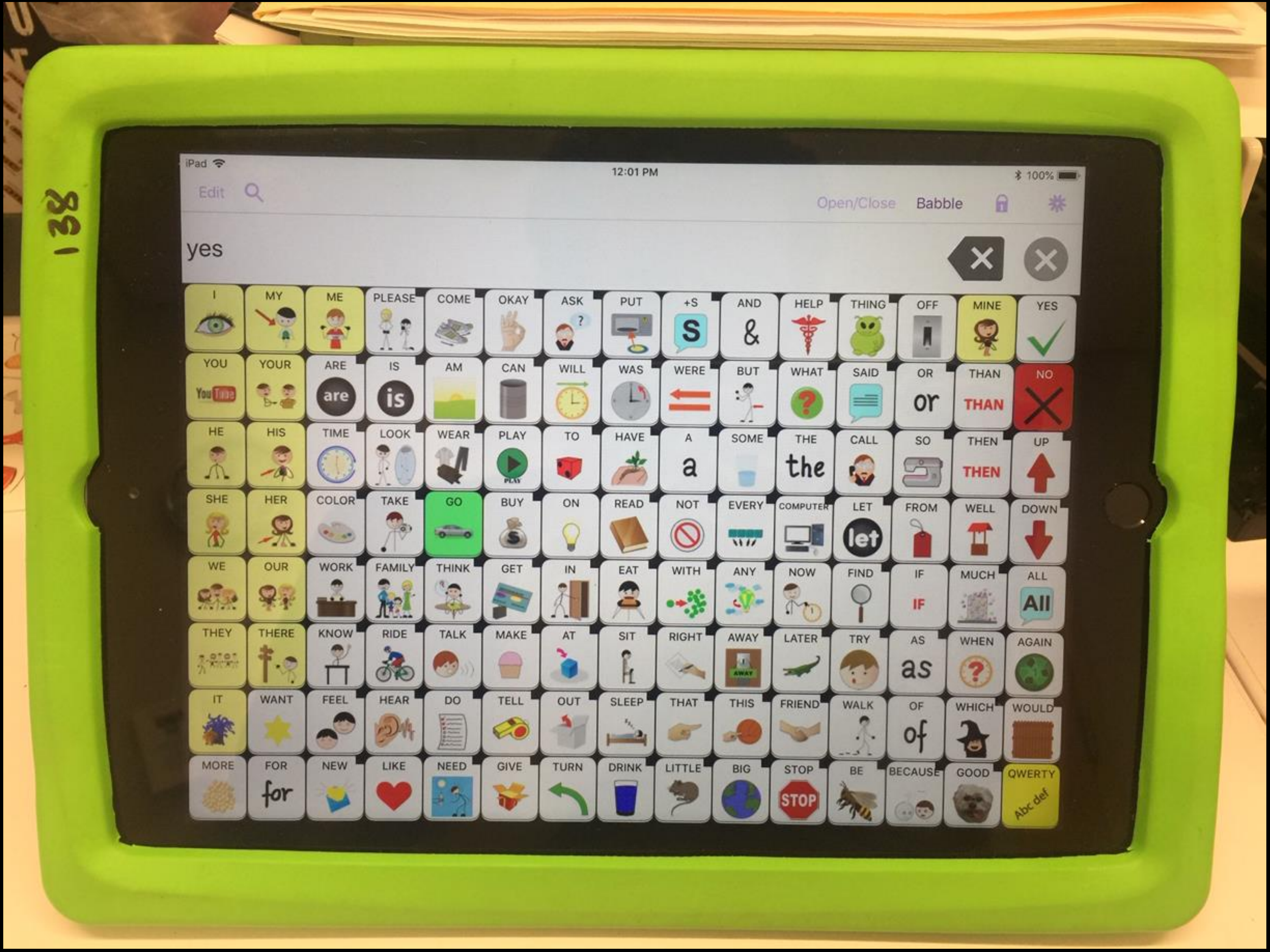
Overall display

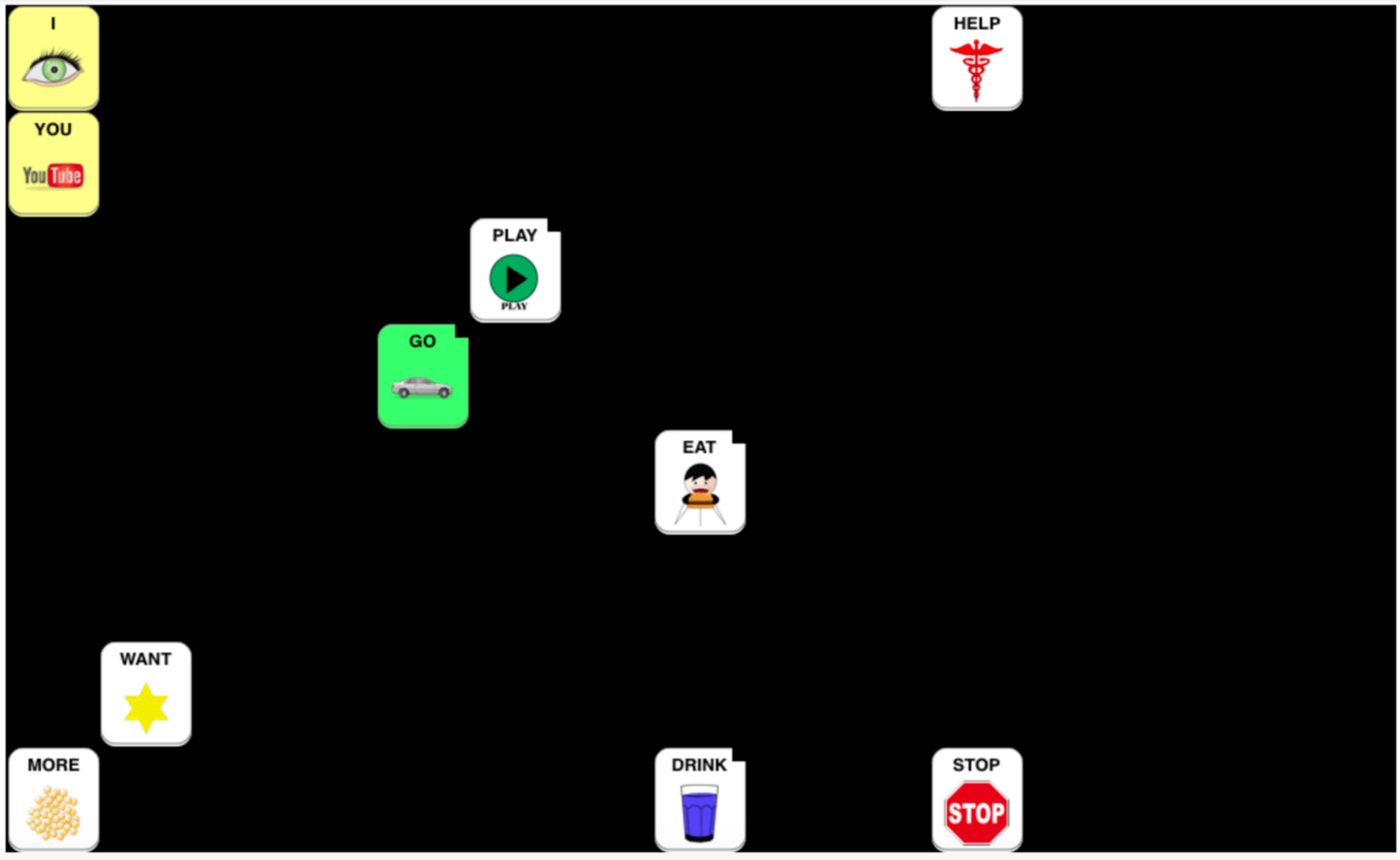
Spacing of array

Individual symbols

Presentation and use

Speak For Yourself (full array)

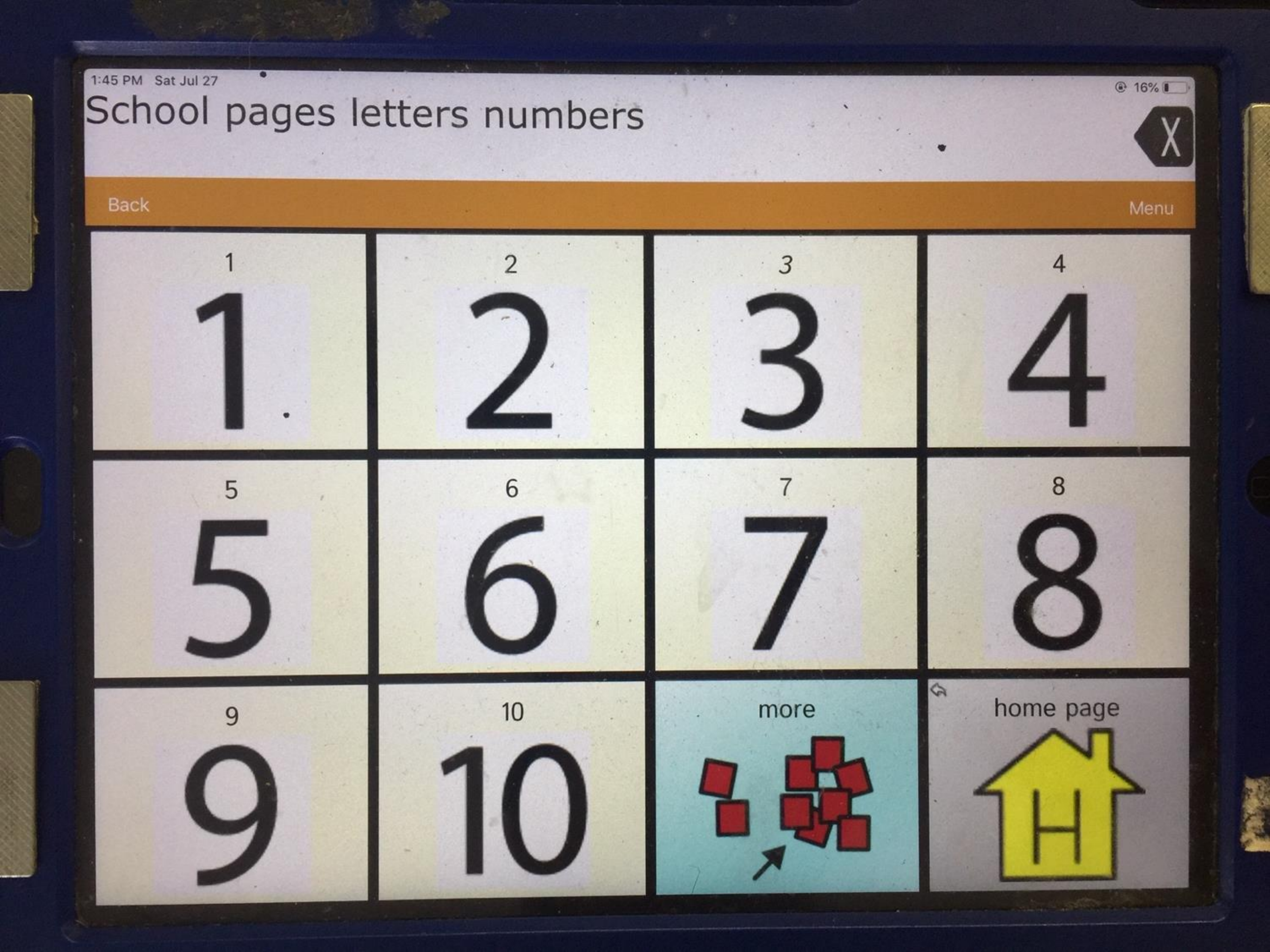


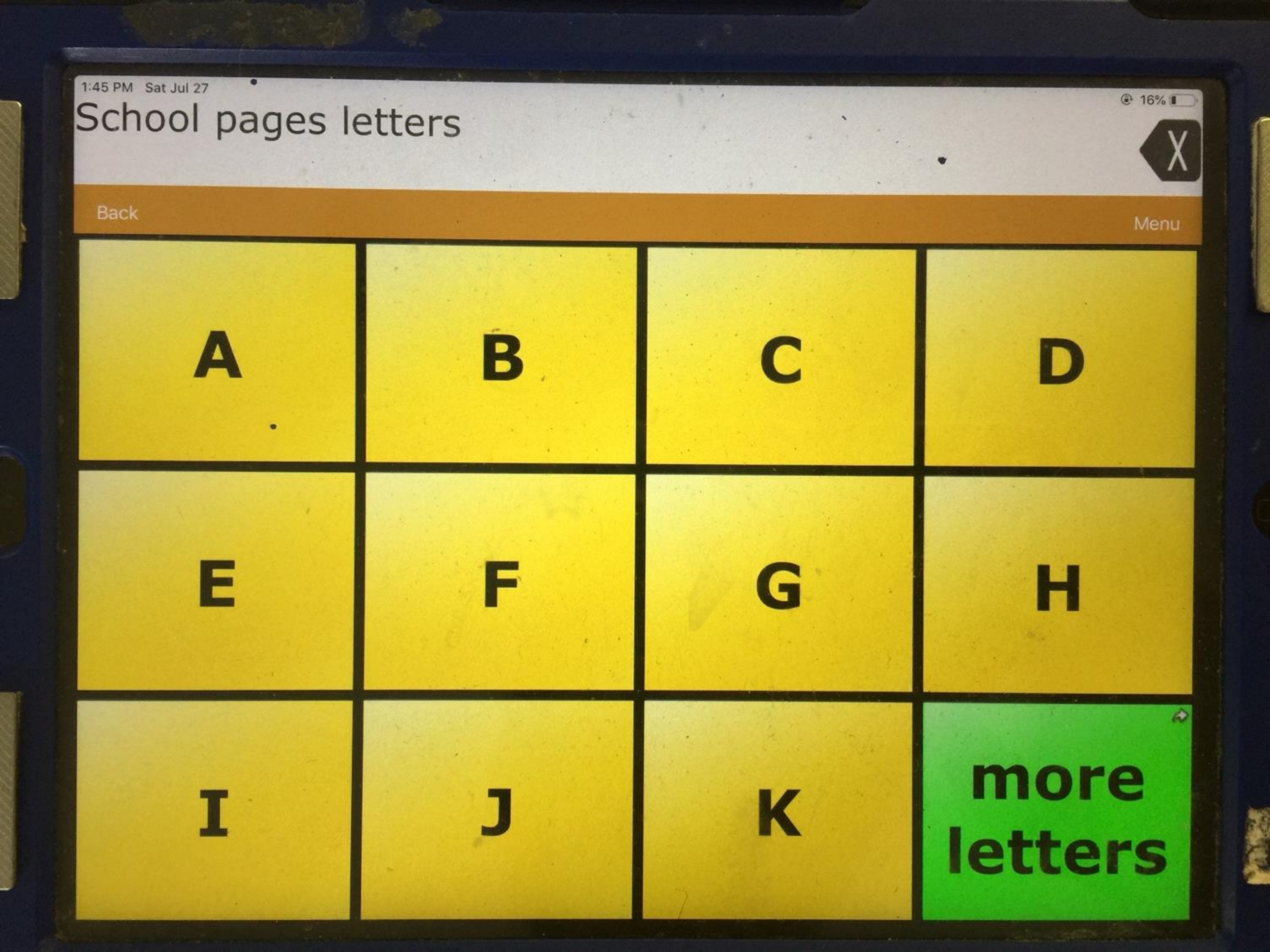


Speak For Yourself (beginning word set)

TouchChat (example)


1:45 PM Sat Jul 27
16%
X
Vocab
Menu
school pages
GOODBYE
HELLO
I ‘m ready
I Want
GO
HELP
NO
stop
STOP
that
YES


1:45 PM Sat Jul 27
16%
School pages letters numbers
X
Back
Menu
1
1
2
2
3
3
4
4
5
5
6
6
7
7
8
8
9
9
10
10
more
home page
H

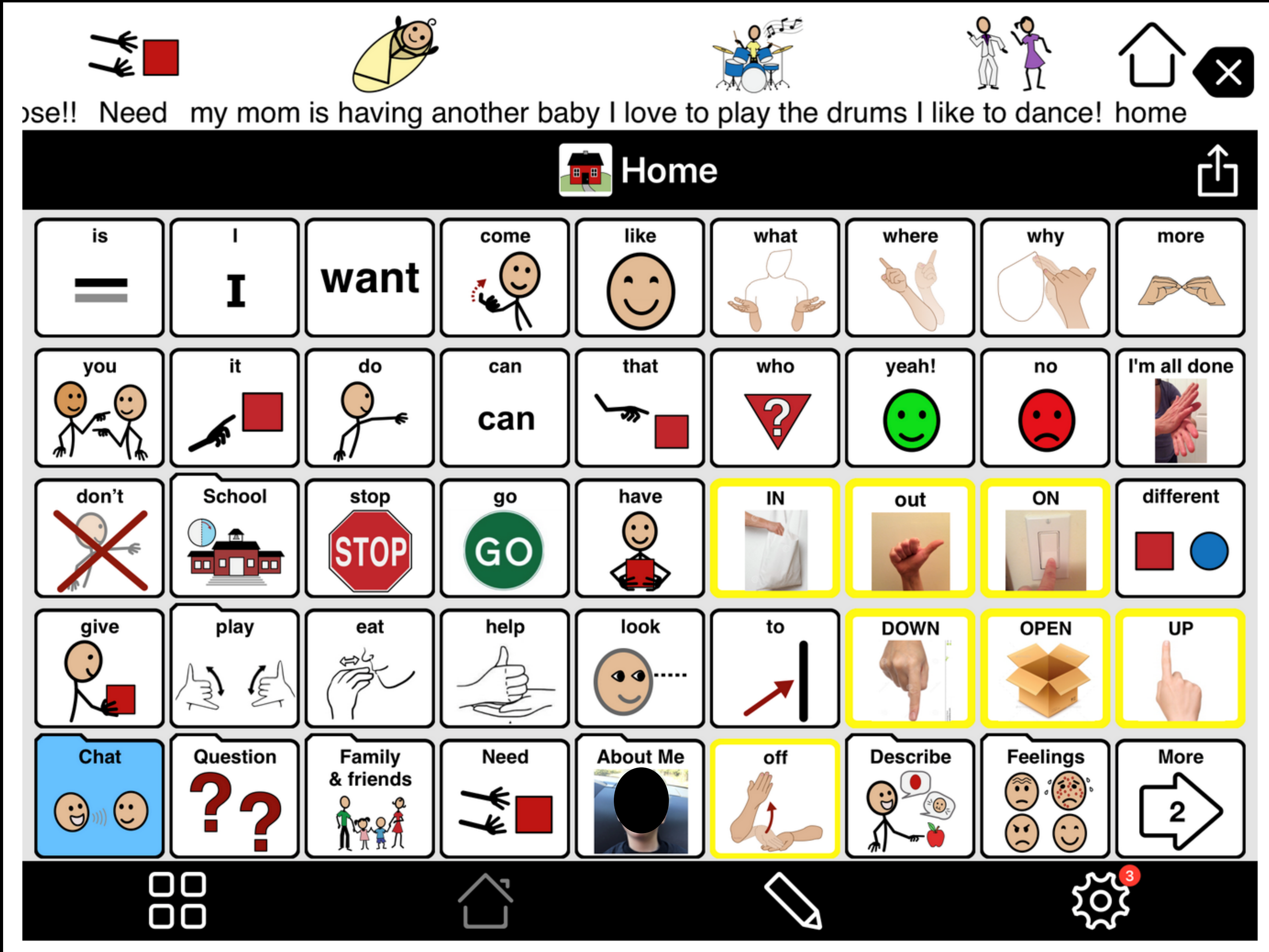

1:45 PM Sat Jul 27
16%
School pages letters
X
Back
Menu
A
B
C
D
E
F
G
H
I
J
K
more letters

Stages in the development of a display…

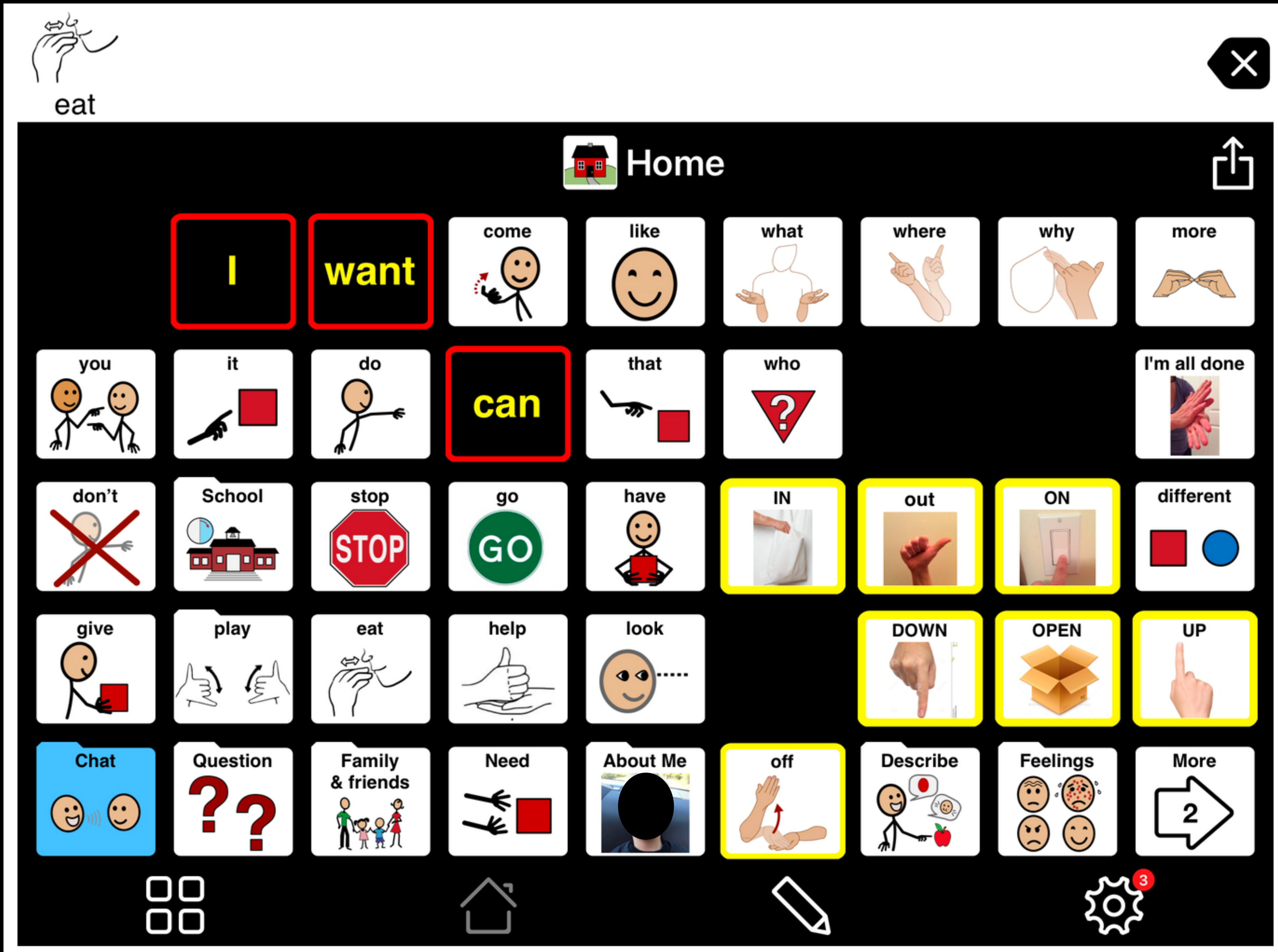
Visual complexity, conceptual complexity:

- Photos
- Line drawings, abstract 2D symbols
- Color highlights to salient features
- Sight words (bubbled, not)

Is the student expected to access the display **visually** throughout the day, or are other channels being used (motor memory, auditory scanning, combination)?



ɔse!! Need my mom is having another baby I love to play the drums I like to dance! home
Home
is
I
I
want
come
like
what
where
why
more
you
it
do
can
can
that
who
yeah!
no
I'm all done
don't
School
stop
STOP
go
GO
have
IN
out
ON
different
give
play
eat
help
look
to
DOWN
OPEN
UP
Chat
Question
Family & friends
Need
About Me
off
Describe
Feelings
More
2
3



eat
Home
I
want
come
like
what
where
why
more
you
it
do
can
that
who
I'm all done
don't
School
stop
STOP
go
GO
have
IN
out
ON
different
give
play
eat
help
look
DOWN
OPEN
UP
Chat
Question
Family & friends
Need
About Me
off
Describe
Feelings
More
2
3

Keyguards?

Colored Keyguard
http://www.laseredpics.biz

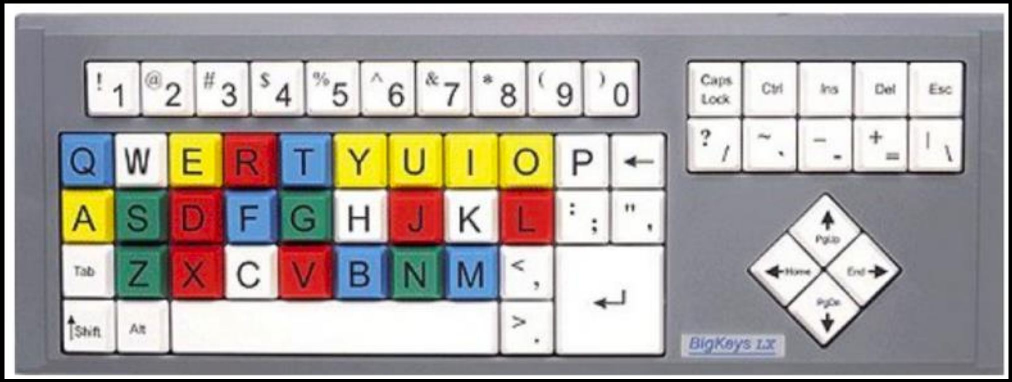
Pros

- Emphasize spacing between symbols
- Color outlines
- Tactile/3D outline

Cons?

Keyboard Adaptations

BigKeys

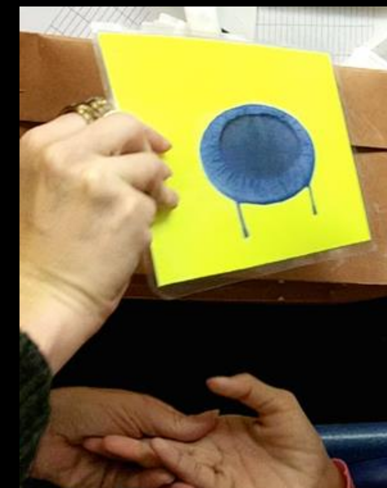


Pimpmykeyboard.com

Remember, it’s not just about material adaptations… Consider:

- Placement and presentation of materials
- Positioning of the student
- Complexity of the environment

Environmental Considerations & Presentation



Complexity

Visual Fields

Movement

Visual Novelty

Light

Distance

Color

Beware of glare!

Positioning and Adaptations for Students with Cortical Visual Impairment

Chart for Planning Activities

Name: | ***Date:***

Current CVI Range Score:

Activity	**CVI Characteristics Impacted** *Note specific impact of activity on multisensory complexity	**Ideal Position of Student** 1. Seated, standing, moving/walking, side-lying, supine, prone, kneeling, cross-legged, etc. 2. Position in room (with respect to lights, windows, and busy areas; visual field preferences)	**Physical adaptive equipment needed** (stander, assistive mobility device, adapted chair or attachment, pillow, tumble form, arm rest, cane, wheelchair, etc.)	**Visual adaptations and accommodations needed** (slant board, black backdrop, reduced complexity of array, color outlines/anchors, flashlight, occluder, movement of materials)

Balancing Sensory Access

- Limit visual demands when focusing on auditory access
  - Literacy example, mid-Phase II: Sight word instruction in short, intense intervals; Reading comprehension with audio books at grade level

This requires planning…

CVI Schedule, CVI Learning Media Profile

Visual & Tactile Modifications to Sign

Phase I:
- Tactile Sign / Hand-Under-Hand Modeling

Phase II (Early → Late):
- Visual sign at near, tactile supports still needed
- Consider visual field preferences - tracking
- Reduced complexity of backdrop is critical
- Allow for "looking away" in visually guided reach

Phase III:
- Visual sign may be accessible at increased distance
- Tactile modifications still may be needed in complex or novel environments

Accessible Back-channeling:
Visual
Tactile
Auditory
Combination/TOTAL COMM.

All Phases: Consider background complexity (clothing, backdrop)

Videos & Resources

Multi-Modal Communication

- Body Language
- Choice Making
- Communication Boards
- Encoding
- Eye Gaze
- Eye Gaze Boards (e-tran)
- Flipbooks
- Gestures
- Partner Assisted Scanning (Live Voice)
- Speech Generating Devices (SGD)
- Spelling (Alphabet Based Strategies)
- Verbal Approximations

THE BRIDGE SCHOOL

http://www.bridgeschool.org/transition/multimodal/eye_gaze_boards.php

Thank you!

Chris Russell
Christopher.russell@qc.cuny.edu

Jennifer Willis
jcwillis@umd.edu